

Networking

LAN Layout

```
T-Mobile KVD21 Gateway (192.168.12.1) – DHCP server, CGNAT, locked DNS, no
settings
└─ Archer AX55 Pro (192.168.12.234) – AP mode (WiFi + switch only, no
routing)
    └─ Tiamat (Proxmox VE)          → 192.168.12.242 (static)
    └─ Bahamut (Raspberry Pi 4)    → 192.168.12.244 (static, DietPi edge
node)
    └─ Laptop                      → 192.168.12.172 (WiFi) / .204 (Ethernet)
    └─ Fire TV ×2                  → DHCP
    └─ Phones ×2                   → DHCP
```

Why AP mode (not Router mode): The T-Mobile KVD21 gateway blocks internet forwarding when a second router does NAT behind it — even the Archer itself cannot ping 1.1.1.1 in Router mode. The KVD21 admin UI has no configurable settings (DNS, DHCP, port forwarding are all locked). AP mode is the only viable configuration.

Archer AP admin: `http://192.168.12.234` (static IP set in AP mode LAN settings) KVD21 admin: `http://192.168.12.1` (read-only status only — no configurable settings)

DNS note: KVD21's DHCP pushes its own DNS (not AdGuard). Set AdGuard DNS manually per-device per `docs/ADGUARD-DEVICE-SETUP.md`. Critical infrastructure (Tiamat, Bahamut, CTs, Home Assistant) already has static DNS pointing to `192.168.12.244`.

"Ziggy" is now CT-900 on Tiamat (Open WebUI + SearXNG). The name no longer refers to a Raspberry Pi.

OpenWrt VM Router

VM-100 `openwrt` is deployed on Tiamat as the future routing/DHCP/DNS control plane.

Current state:

- VMID: `100`
- Storage: `local-ssd`
- WAN NIC: `eth0` on `vmbr0`, DHCP from KVD21/Archer LAN (`192.168.12.x` ; currently observed as `192.168.12.145`)
- LAN NIC: `eth1` on `vmbr1`, static `10.10.0.1/24`

- LuCI: `http://10.10.0.1/` from Tiamat or any host attached to `vmbr1`
- SSH: `ssh root@10.10.0.1` from Tiamat
- Root password: stored on Tiamat at `/root/openwrt-root-password` with mode `0600`
- Package manager: `apk` on OpenWrt 25.12.2, not `opkg`
- Proxmox guest agent: enabled and verified with `qm agent 100 ping`

Important limitation: because `vmbr1` is virtual-only, OpenWrt currently serves only virtual clients attached to `vmbr1`. It does not serve phones, Fire TVs, or laptop WiFi yet. Full physical-device DHCP/DNS requires the TP-Link UE300 USB-A gigabit NIC documented in `docs/HARDWARE.md`; that NIC will become a physical OpenWrt LAN bridge (`vmbr2`) connected to the Archer AP/switch.

Management commands from Tiamat:

```
qm terminal 100
ssh root@10.10.0.1
qm agent 100 ping
```

Setup script: `proxmox/setup-openwrt-vm.sh` creates `vmbr1`, imports the OpenWrt x86/64 EFI image, configures WAN/LAN assignment, installs LuCI and qemu guest agent, and verifies the VM.

Proxmox Bridge

Proxmox host networking is bridged through `vmbr0` so all existing CTs/VMs get LAN access. `vmbr1` is a virtual-only internal bridge for the OpenWrt VM LAN; it has no physical port until the TP-Link UE300 USB NIC arrives.

`/etc/network/interfaces` :

```
auto lo
iface lo inet loopback

auto enp4s0
iface enp4s0 inet manual

auto vmbr0
iface vmbr0 inet static
    address 192.168.12.242/24
    gateway 192.168.12.1
    bridge-ports enp4s0
    bridge-stp off
    bridge-fd 0
    dns-nameservers 192.168.12.244

auto vmbr1
iface vmbr1 inet manual
    bridge-ports none
    bridge-stp off
    bridge-fd 0
    bridge-multicast-snooping 0
```

Core Service Map

Consolidated 2026-05-22: All Proxmox LXC services merged into CT-300. **2026-05-28:** CT-501 `habridge` added for Alexa/Hue voice control.

Proxmox host

Service	URL
Proxmox UI	<code>https://192.168.12.242:8006</code>

HABridge — Alexa voice bridge (CT-501, port 80)

Service	Port	URL
HABridge web UI + Hue API	80	<code>http://192.168.12.251 / habridge.tiamat.local</code>
UPnP/SSDP discovery	1900/udp	(Alexa auto-discovers on LAN)

CT-300 `mediastack` (192.168.12.30)

All services run inside this single privileged Debian 12 LXC.

Service	Internal port	URL / hostname
Riven frontend	3000	<code>http://192.168.12.30:3000 / riven.mediastack.lan</code>
Riven backend	8080	<code>http://192.168.12.30:8080</code>
Jellyfin	8096	<code>http://192.168.12.30:8096 / jellyfin.mediastack.lan / jellyfin.tiamat.local</code>
n8n	5678	<code>http://192.168.12.30:5678 / n8n.tiamat.local</code>
Threadfin (IPTV proxy)	34400	<code>http://192.168.12.30:34400/web / threadfin.tiamat.local</code>
Uptime Kuma	3001	<code>http://192.168.12.30:3001 / uptime.tiamat.local</code>
Homarr (dashboard)	7575	<code>http://192.168.12.30:7575 / homarr.mediastack.lan · board: /boards/Tiamat-Mediastack · login: admin / Mediastack2024!</code>
Pulse (Proxmox monitor)	—	web UI port in <code>/etc/pulse/.env</code>
Caddy (reverse proxy)	80/443	handles all <code>*.tiamat.local</code> + <code>*.mediastack.lan</code> routes
PostgreSQL 15	5432	internal only (<code>riven</code> database)
Redis	6379	internal only
CrowdSec	8081	internal only
Tailscale	—	node <code>tiamat-tailscale</code> · <code>100.115.82.71</code> · funnel <code>https://tiamat-tailscale.tail9d8b73.ts.net</code>

Decommissioned CTs (2026-05-22): CT-102 (n8n-stable), CT-103 (Traefik), CT-104 (Vaultwarden), CT-105 (Valkey), CT-106 (Postgres), CT-107 (Authentik), CT-110 (Pulse), CT-200 (HA DuckDNS proxy), CT-234 (Threadfin), CT-248 (Uptime Kuma), CT-279 (Tailscale)

Running CTs (2026-05-28): CT-300 `mediastack` (192.168.12.30), CT-501 `habridge` (192.168.12.251) Vaultwarden backup: `/var/backup/mediastack/vaultwarden_pg_dump.sql` + `vaultwarden_data.tgz` in CT-300.

VMs (unchanged)

VM	Purpose	IP	Notes
VM-100 <code>openwrt</code>	OpenWrt router/control plane	<code>10.10.0.1</code> LAN / <code>192.168.12.x</code> WAN	WAN on <code>vbr0</code> , LAN on virtual <code>vbr1</code> , LuCI + qemu-ga enabled
VM-901 <code>windows-gaming</code>	Windows 11 gaming	<code>192.168.12.201</code>	RX 580 GPU passthrough
VM-990 <code>haos17-1</code>	Home Assistant OS	<code>192.168.12.123</code>	Smart home hub —  static IP set via <code>ha network update enp6s18 --ipv4-method static</code> (permanent, no DHCP needed)

Laptop NFS Exports (192.168.12.172)

Export path	Tiamat mount	Consumer
/media/loufogle/Data/Calibre Library	/mnt/laptop/calibre	CT-233 Calibre-Web
/media/loufogle/Data/Cookbooks	/mnt/laptop/cookbooks	CT-233 second library
/media/loufogle/SystemBackup/Videos	/mnt/laptop/videos	Plex / Jellyfin Home Videos
/media/loufogle/ISOs1	/mnt/laptop/isos	Proxmox ISO uploads
/media/loufogle/Games/roms	/mnt/laptop/roms	Future emulation frontend
/media/loufogle/Data/Downloads/lou.m3u	/mnt/laptop/iptv/lou.m3u	Jellyfin Live TV / TVHeadend CT-235

See `docs/NFS.md` for setup.

VPN Architecture

Active (2026-06-06). PiVPN + WireGuard on Bahamut. All systems behind VPN.

Server: Bahamut (`192.168.12.244:51820`), PiVPN, subnet `10.92.29.0/24` . IP forwarding + NAT masquerade on Bahamut forwards all client traffic.

Client	VPN IP	Config	Boot
Laptop	10.92.29.2	<code>/etc/wireguard/wg0.conf</code>	<code>systemctl enable wg-quick@wg0</code>
Tiamat	10.92.29.3	<code>/etc/wireguard/wg0.conf</code>	<code>systemctl enable wg-quick@wg0</code>
OpenWrt VM-100	10.92.29.4	UCI <code>network.wg0</code>	persistent (UCI)
CT-300 mediastack	10.92.29.5	<code>/etc/wireguard/wg0.conf</code>	<code>systemctl enable wg-quick@wg0</code>

All clients use full tunnel (`AllowedIPs = 0.0.0.0/0`) with `Endpoint = 192.168.12.244:51820` . DNS through VPN: `10.92.29.1` (Bahamut WG interface → AdGuard Home).

Manage clients: `ssh bahamut "pivpn -l" / pivpn add -n <name> / pivpn -r <name>` . Tray widget on laptop: `~/wireguard-tools/tray-widget/wireguard-tray-widget.py` .

Old architecture (decommissioned): CT-100 WG server, CT-101 WG proxy + TinyProxy, wg-easy Docker on Bahamut. Preserved in `infrastructure/wireguard-server/` for reference.

aria2 Fallback Download Path

When Real-Debrid refuses a torrent (DMCA, dead, uncached), aria2 in CT-300 downloads it directly via BitTorrent through the WireGuard tunnel.

Component	Location	Port
aria2 daemon	CT-300	6800 (JSON-RPC)
riven-aria2-bridge	CT-300	— (polls Riven API)
AriaNg (optional)	—	—

Config: `/etc/aria2/aria2.conf` , RPC secret: `mediastack-aria2` . Bridge: `/opt/riven-aria2-bridge.py` , polls Riven Failed items every 60s. State: `/var/lib/riven-aria2-bridge/state.db` . Downloads to: `/data/media/downloads/` , then Jellyfin picks up via library scan.

Caddy Reverse Proxy (CT-300)

All *.tiamat.local and *.mediastack.lan hostnames are served by Caddy inside CT-300 at 192.168.12.30:80/443 . DNS wildcard: *.tiamat.local → 192.168.12.30 via AdGuard rewrite on Bahamut. Config: infrastructure/caddy/Caddyfile (also live at /etc/caddy/Caddyfile in CT-300).

Hostname	Service	Backend
riven.mediastack.lan , riven.local	Riven UI	127.0.0.1:3000
jellyfin.mediastack.lan , jellyfin.local	Jellyfin	127.0.0.1:8096
n8n.tiamat.local	n8n	127.0.0.1:5678
uptime.tiamat.local , uptimekuma.tiamat.local	Uptime Kuma	127.0.0.1:3001
threadfin.tiamat.local	Threadfin	192.168.12.30:34400
ha.tiamat.local, homeassistant.tiamat.local	Home Assistant VM-990	192.168.12.123:8123
habridge.tiamat.local	HABridge CT-501	192.168.12.251:80
sonarr.tiamat.local	Sonarr (stopped CT-214)	192.168.12.214:8989
radarr.tiamat.local	Radarr (stopped CT-215)	192.168.12.225:7878
prowlarr.tiamat.local	Prowlarr (stopped CT-210)	192.168.12.210:9696
homarr.mediastack.lan , homarr.local	Homarr dashboard	127.0.0.1:7575
guide.mediastack.lan , guide.local	Guide / Meiliseach	127.0.0.1:7700
cockpit.mediastack.lan	Cockpit	127.0.0.1:9090

To add a route, append a block to /etc/caddy/Caddyfile in CT-300 and systemctl reload caddy . Also sync to infrastructure/caddy/Caddyfile in this repo.

CT-103 Traefik decommissioned 2026-05-22. Old Traefik configs preserved in infrastructure/traefik/dynamic/ for reference.

DNS / Remote Access on Bahamut

Bahamut (192.168.12.244 , Raspberry Pi 4, DietPi / Debian 13) is the native edge-services node. It uses fish as root's login shell.

Native services:

- AdGuard Home: DNS :53 , admin UI http://192.168.12.244:8081
 - Rewrite: *.tiamat.local → 192.168.12.30 (CT-300 Caddy)
 - Required explicit rewrites: habridge.tiamat.local → 192.168.12.30 , ha.tiamat.local → 192.168.12.30
- Caddy + DuckDNS: native /usr/local/bin/caddy , systemd caddy.service , TLS for *.lou-fogle-media-stack.duckdns.org
 - vaultwarden.* → localhost:8080
 - adguard.* → localhost:8081
 - ha.* → 192.168.12.123:8123 (Home Assistant VM-990)
- Vaultwarden: native ARM64 binary built on the laptop and deployed to /usr/local/bin/vaultwarden , systemd vaultwarden.service , data in /opt/appdata/vaultwarden
- DietPi Dashboard: :5252
- TigerVNC: :5901
- PiVPN + WireGuard: VPN server :51820/udp , subnet 10.92.29.0/24
 - Clients: laptop, tiamat, openwrt, mediastack (see VPN Architecture)

Removed from active Bahamut architecture:

- Docker / containerd
- wg-easy (replaced by native PiVPN)
- Docker Caddy image

Tailscale removed from Bahamut (2026-05-22). Tailscale now runs inside CT-300 as tiamat-tailscale (100.115.82.71).

Router DNS recommendation:

1. 192.168.12.244

Device Inventory

Captured 2026-06-27 via `nmap -sn 192.168.12.0/24` from the laptop, with device labels confirmed by the operator. Phones/TVs/tablets/Echos use DHCP, so the **MAC is the stable identifier** — IPs here are reference only. File shares filter by subnet + auth (not per-device IP), so a changing DHCP lease never breaks access.

Infrastructure (static)

IP	Device	MAC	Notes
192.168.12.1	T-Mobile KVD21 gateway	C8:99:B2:E1:82:C9	ISP gateway, locked
192.168.12.234	TP-Link Archer AX55 Pro	24:2F:D0:28:E0:05	AP mode
192.168.12.242	Tiamat (Proxmox host)	B4:2E:99:AA:CF:06	Gigabyte; file-share hub
192.168.12.244	Bahamut (Pi 4, eth0)	88:A2:9E:85:CD:E6	AdGuard DNS
192.168.12.245	Bahamut (Pi 4, wlan0)	88:A2:9E:85:CD:E7	secondary iface
192.168.12.30	CT-300 mediastack	BC:24:11:E3:7C:DB	Jellyfin/Riven/n8n
192.168.12.123	VM-990 Home Assistant OS	02:88:F9:77:28:E2	
192.168.12.145	VM-100 OpenWrt	BC:24:11:BB:18:00	
192.168.12.251	CT-501 HABridge	BC:24:11:AB:A3:2D	
192.168.12.172	Laptop (Pop!_OS, WiFi)	—	pop-os.lan
192.168.12.204	Laptop (Pop!_OS, Ethernet)	—	control center
192.168.12.215	HDHomeRun tuner	00:18:DD:04:8E:EE	HDHR-1048EEE4
192.168.12.198	Epson printer	68:55:D4:34:0E:C8	EPSON340EC8

Endpoints (DHCP)

IP	Device	MAC	Notes
192.168.12.112	Fire TV — living room	78:A0:3F:2D:B9:E0	
192.168.12.188	Fire TV — bedroom	D8:E7:43:20:7C:64	
192.168.12.222	Echo Show	A4:08:01:0A:2B:99	Amazon; which Echo is .222 vs .100 to confirm
192.168.12.100	Echo — bedside	0C:DC:91:35:33:8B	Amazon
192.168.12.177	Galaxy Note 20 (phone)	—	offline at scan
192.168.12.120	Android tablet	—	offline at scan
(DHCP, TBD)	Galaxy S10 (2nd phone)	—	offline at scan

Unidentified (investigate)

IP	MAC	Vendor	Notes
192.168.12.140	5C:E7:53:45:EA:3E	Shenzhen Intellirocks	unknown IoT/smart-home device

Re-run discovery any time: `sudo nmap -sn 192.168.12.0/24` from the laptop; resolve a MAC vendor with `curl https://api.macvendors.com/<MAC>` .